Summary in Graph

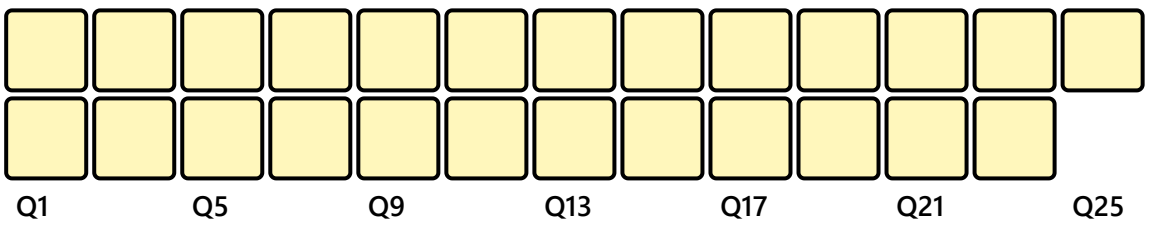
Exam Summary (GO Classes CS Test Series 2026 | CO and Architecture | Subject Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	25 (5 + 20)
Total Marks:	45 (5 + 40)
Exam Duration:	75 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (16) out of 161 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Numerical Type Award: 1 Penalty: 0 CO & Architecture

What is the **minimum number of registers** needed to perform register allocation for the code shown below:

```
t1 := a - mem[...]
t2 := d / mem[...]
t3 := t2 + mem[...]
t4 := t1 + t3
```

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

The performance of a pipelined processor suffers if:

A. The pipeline stages have different delays.

B. Consecutive instructions are dependent on each other.

C. The pipeline stages share hardware resources.

D. All of the above.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Analyze the following C code fragment:

```
A[12] = h + a[8]
```

In MIPS, which assembly code corresponds to it?

A.
```
lw $t1, 12($s3)
add $t0, $s2, $t0
sw $t0, 24($s3)
```

B.
```
lw $t0, 32($s3)
add $t0, $s2, $t0
sw $t0, 48($s3)
```

C.
```
lw $t0, 6($s3)
add $t0, $s2, $t0
sw $t1, 12($s3)
```

D.
```
lw $t1, 32($s3)
add $t0, $s2, $t0
sw $t1, 48($s1)
```

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

About cache memory, consider the following statements.

I. In associative mapping, each main memory block can be loaded into any cache line.

II. In direct mapping, each main memory block is mapped to only one cache line.

III. In direct mapping, repeated access to different memory blocks mapped to the same cache line will result in a high hit rate.

IV. The associative mapping technique is simple and inexpensive to implement.

Mark the correct option.

A. Only statements I and II are correct.

B. Only statements I and IV are correct.

C. Only statements III and IV are correct.

D. Only statements I, II and III are correct.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

Most relevant addressing mode to write position independent code is

A. Direct

B. Indirect

C. Relative

D. Indexed Mode

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

The interposition of a cache memory circuit between the processor and the main memory (RAM)

A. increases the traffic of instructions and/or data on the memory bus

B. decreases the traffic of instructions and/or data between memory and disk

C. decreases the traffic of instructions and/or data on the memory bus

D. decreases the traffic of instructions and/or data between memory and disk

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a CPU using a $5-$stage pipeline structure (IF, ID, EX, MEM, WB), with separate data and instruction memories, no data forwarding mechanism, writing to the register file on the rising edge of the clock and reading on the falling edge of the clock, and the following set of instructions:

$$\begin{array}{lll} \text{I1:} & \text{lw} & \$2,\ 100(\$5) \\ \text{I2:} & \text{add} & \$1,\ \$2,\ \$3 \\ \text{I3:} & \text{sub} & \$3,\ \$2,\ \$1 \\ \text{I4:} & \text{sw} & \$2,\ 50(\$1) \\ \text{I5:} & \text{add} & \$2,\ \$3,\ \$3 \\ \text{I6:} & \text{sub} & \$2,\ \$2,\ \$4 \end{array}$$

How many clock cycles are spent executing this code?

Your Answer: | Correct Answer: 17 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Regarding the following statements about cache memory, which are true?

I. In a fully associative structure, a memory block can be mapped to any cache slot.

II. The tag field of the address is used to identify a valid block in the cache, together with the index field.

III. A $\text{level } 2$ cache serves to reduce the penalty in case of a miss at $\text{level } 1$.

IV. The LRU replacement scheme is the most used for the direct-mapped structure

A. Only statements I, III and IV.

B. Only statements II, III and IV.

C. Only statements I and II.

D. Only statements I, II and III.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Assume a program has an execution profile in which 85% of the instructions are simple (such as **AND, XOR, ADD, and BRANCH**) and the remaining 15% are complex instructions (such as **MUL and DIV**). Also assume that simple instructions require 2 **machine cycles** and complex instructions require 12 **cycles** on a **CISC** machine (each cycle $= 10\ \text{ns}$). On a **RISC** machine, simple instructions are executed in 1 **cycle**, while complex instructions must be simulated by software, requiring on average 20 **cycles per instruction.** Because of its simplicity, the cycle time on a RISC machine is $8\ \text{ns}$. Also consider that the program required $100,000,000$ **instructions** to complete.

What is the time spent (in seconds) to execute this program, respectively, on the **CISC** and **RISC** machines?

A. 1 and 0.8

B. 3.5 and 3.08

C. 10 and 12.8

D. 8.5 and 1.5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Analyze the following statements.

I. The processor that presents the best performance is always the one that has the highest clock frequency.

II. The pipelining technique is used to increase performance in processors. In this way, the pipeline reduces the instruction latency time.

III. The simplest way to increase the hit rate in cache memory is to increase its capacity.

IV. In superscalar architectures, the effects of data dependencies and anti-dependencies are reduced in the register renaming stage.

From the analysis, it can be concluded that:

A. all statements are true.

B. only statements II and III are true.

C. only statements III and IV are true.

D. none of the statements is true.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A computer has a memory system organized into four levels: $\text{level } 1$ and $\text{level } 2$ cache memories, main RAM memory, and secondary memory. Programs ready for execution are brought from secondary memory and transformed into processes in main memory. The instruction used to access data in memory provides the **real memory address** where the desired information is located. The information is then searched in $\text{level } 1$ cache. If it is not found, it is searched in $\text{level } 2$ cache. If still not found, the information is finally fetched from main memory.
What addressing mode is used?

A. Immediate

B. Indirect

C. Direct

D. Implicit

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A RISC processor is implemented in two synchronous organization versions: a **single-cycle** one, in which each instruction executes in exactly one clock cycle, and a $5-$**stage pipeline** version. The stages of the pipeline version are:

1. Instruction fetch,
2. Operand fetch,
3. Operation execution,
4. Memory access, and
5. Register file update.

The maximum operating frequency of the organizations is calculated as $100\ \mathrm{MHz}$ for the single-cycle version and $400\ \mathrm{MHz}$ for the pipeline version. A program X that executes 200 instructions is used to compare the performance of the organizations. Of the 200 **instructions**, only 40% access memory, while the others operate only on internal registers of the organization. Assume the program presents **no data or control conflicts** between instructions that may be simultaneously inside the pipeline of the second organization. Thus, the execution time of program X in the single-cycle and pipeline organizations is, respectively:

A. 2000 nanoseconds and 510 nanoseconds.

B. 2000 nanoseconds and 500 nanoseconds.

C. 2000 nanoseconds and 2300 nanoseconds.

D. 2300 nanoseconds and 500 nanoseconds.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

A computer system has a 4 **Gbyte** memory map, using **byte addressing**, and a cache memory with a **direct-mapped** organization. The cache can store up to $1{,}024$ **words of** 32 **bits** from the memory map. Assume the cache is always written **atomically with four bytes** coming from a memory address aligned on a $32-$**bit word boundary**, and that it uses 1 **valid bit per cache line.**

In this case, the sizes of the cache **tag, index,** and the **cache size**, respectively, are:

A. 12 bits, 18 bits and $54,272$ bits.

B. 14 bits, 18 bits and $56,320$ bits.

C. 20 bits, 10 bits and $54,272$ bits.

D. 20 bits, 12 bits and $54,272$ bits.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A program $\mathbf{P}$ is executed on a computer $\mathbf{C}$, which has a clock speed of $\mathbf{2\ GHz}$, in $\mathbf{5}$ **seconds.** It is desired to design a computer $\mathbf{C2}$ such that the same program $\mathbf{P}$ is executed in $\mathbf{3}$ **seconds** on $\mathbf{C2}$. In the design, it was found that to achieve this goal, program $\mathbf{P}$ will spend $\mathbf{50\%}$ more clock cycles on $\mathbf{C2}$ than on $\mathbf{C1}$.

The **frequency**(in GHz) that $\mathbf{C2}$ must have so that $\mathbf{P}$ is executed in $\mathbf{3}$ **seconds**?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Regarding **instruction sets**, mark **T (true)** or **F (false)** for the statements below.

- The opcode specifies the operation to be performed.
- Operand references specify a register or memory.
- The little-endian style stores the most significant byte at the lowest address.
- Stacks may or may not be visible to the programmer.
- Stacks grow from lower addresses to higher addresses.

Choose the option that contains, from **top to bottom**, the correct sequence.

A. $\mathrm{T, T, F, T, F.}$

B. $\mathrm{T, F, T, F, F.}$

C. $\mathrm{F, T, T, T, F.}$

D. $\mathrm{F, T, F, F, T.}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider the following MIPS Assembly program fragment:

```
.data 0x10010000  # data segment
palavra1: .word 13
palavra2: .word 0x15
```

In hexadecimal, the addresses of **`palavra1`** and **`palavra2`** are, respectively:

A. `0x10010002` and `0x10010002`

B. `0x10010000` and `0x10010004`

C. `0x10010002` and `0x10010006`

D. `0x10010000` and `0x10010008`

(Here, **`.word`** allocates 4 bytes for each variable.)

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #17 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A computer has a two-level cache. Suppose that 80% of the references to information are serviced by the $\textbf{level-1 cache}$, 15% by the $\textbf{level-2 cache}$, and 5% by **main memory.** The access times are $\mathbf{5\ ns}, \mathbf{15\ ns}$, and $\mathbf{60\ ns}$, respectively. What is the **average access time** of the system (in ns)?

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Using your knowledge about the RISC architecture, analyze the statements below:

1. RISC architecture supports arbitrary data alignment for load/store operations.
2. No indirect addressing that requires a memory access to obtain the address of an operand in memory is considered typical in a classic RISC.
3. In RISC architecture, the number of bits for integer register specifiers is four or more. This means that at least 16 integer registers can be explicitly referenced at a time.
4. A smaller number of addressing modes, normally fewer than five. This parameter is difficult to determine in RISC architecture.
5. RISC architecture supports operations that combine load/store with arithmetic (for example, add from memory, add to memory).
6. No more than one memory-addressed operand per instruction is specific to RISC architecture.
7. In RISC architecture, instruction size is normally 4 bytes.

The result of summing the numbers corresponding to the correct option is:

A. 17

B. 10

C. 18

D. 12

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Instruction-level parallelism exists when the instructions in a sequence are independent and, therefore, can be executed in parallel or overlapped. As an example of the concept of parallelism, consider the two code fragments (A and B) below :

Code A	Code B
Load $R1 \leftarrow R2$	Add $R3 \leftarrow R3,$ "1"
Add $R3 \leftarrow R3,$ "1"	Add $R4 \leftarrow R3,\ R2$
Add $R4 \leftarrow R4,\ R2$	Store [R4] $\leftarrow$ R0

Mark the **INCORRECT** alternative about parallelism.

A. The degree of instruction-level parallelism is determined by the frequency of true data dependencies and procedural dependencies in the code.

B. The three instructions in Code A are independent and, in theory, all could be executed in parallel.

C. The three instructions in Code B cannot be executed in parallel, because the second instruction uses the result of the first and the third uses the result of the second.

D. Machine parallelism is a performance measure, determined by the number of instructions that can be issued and executed at the same time in a sequential manner.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Hit ratio of the cache memory read request is 85% and the cache memory is 5 **times faster** than main memory. Block size in memory organization is **words.** The access time of the main memory is $72\ \text{ns}$ **per word. Write-through** protocol is used in the system. CPU generates 60% of the requests as **read** requests to read the data and the remaining as **write** operations.

What is the **average access time** of the memory considering both **read and write** operations?

A. $89.28\ \text{ns}$

B. $62.2\ \text{ns}$

C. $42.624\ \text{ns}$

D. $72\ \text{ns}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider the program segment consists of 16 instructions (I1, I2, ... I16) used to execute on pipelines A and B. In the program segment all instructions consume same amount of time to execute.

Pipeline A has 6 stages $(\text{S1, S2, S3, S4, S5, S6})$ with the respective stage delays of $2\ \text{ns}, 6\ \text{ns}, 5\ \text{ns}, 2\ \text{ns}, 8\ \text{ns}$ and $1\ \text{ns}$. Pipeline B is designed with same number of stages but stage 5 will be divided into 4 sub stages using the equal delay i.e. $(8\text{ns}/4)$. If the program execution time of pipeline A is x and the program execution time of pipeline B is y. What is the value of (x/y) *(upto two decimal place)*?

Your Answer: | Correct Answer: 1.17 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

A computer uses $\mathbf{8}$ digit mantissa and $\mathbf{2}$ digit exponent. If $\mathbf{a = 0.052}$ and $\mathbf{b = 28E + 11}$ then $\mathbf{b + a - b}$ will be :

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Non pipelined system takes $130\ \mathrm{ns}$ to process an instruction . A program of 1000 instructions is executed in non pipelined system. Then same program is processed with processor with 5 segment pipeline with clock cycle of $30\ \mathrm{ns/stage}$.

Determine speed up ratio of pipeline *(upto two decimal points).*

Your Answer: | Correct Answer: 4.32 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

The range of representable normalized numbers in IEEE-754 floating-point binary fractional representation but with **bias**$= \mathbf{128}$ using a 32-bit word is

$$2^a to \left(2 - 2^b\right) \times 2^c$$

The value of $a + b + c$ is _____ .

Your Answer: | Correct Answer: -24 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider a 2-way set associative cache (initially empty) with a total of 16 cache blocks. The main memory consists of 256 blocks and the requests for memory blocks are in the following order:

$$0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155$$

Which one of the following memory blocks **WILL** be in the cache if an LRU replacement policy is used?

A. 0

B. 8

C. 129

D. 216

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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